

Qatar-1b

R-2766

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140606 · created 2026-09-03T00:52:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456814.8934 +/- 0.00031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.042
Transit depth (Rp/Rs)^2	1.1 +/- 0.87 [%]
Orbital Inclination (inc)	86.29 +/- 2.77
Airmass coefficient 1 (a1)	0.937 +/- 0.024
Airmass coefficient 2 (a2)	0.057 +/- 0.07
Scatter in the residuals of the lightcurve fit is	2.51 %
Best Comparison Star	#2 - [582, 114]
Optimal Method	PSF photometry
Transit Duration (day)	0.07 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.105 $\pm$ 0.042 vs 0.1463 (deviation 0.68 σ)
Duration fit vs published	0.07 d vs 0.0692 d (deviation 0.04 σ)
Mid-time O-C	0.4 min
Depth SNR	1.7
Residual scatter	2.51 %

Light curve

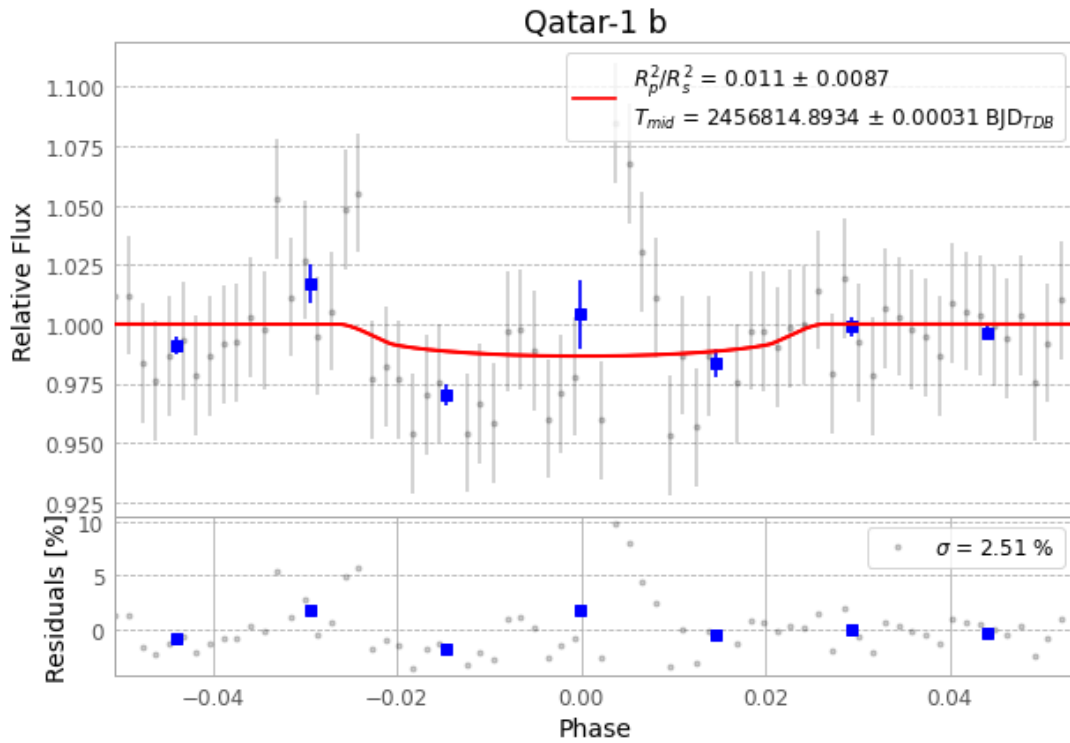


Figure 1 — FinalLightCurve_Qatar-1 b_2014-06-06.png (SHA-256 1d314f012a653c5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6596299509c655a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1140606074212.FITS → Qatar-1140606111513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140606.log (9caf45adb875...), FinalLightCurve_Qatar-1 b_2014-06-06.png (1d314f012a65...), FinalLightCurve_Qatar-1 b_2014-06-06.csv (1af8a8df0772...)

Compute resources

Wall time 136.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:52Z.